Homework 2

Module 2 - IV Bolus Administration · Homework · No date

Name _____ Date _____

1. A 165-lb male received a single IV bolus dose of an antibacterial agent at a dose level of 20 mg/kg. The concentration of drug in the plasma was 113.7 mcg/mL and 49.5 mcg/mL at 2 and 8 hours, respectively, following administration of the dose. The antibacterial agent displays, linear, first-order, one compartment pharmacokinetics.

1a What is the half-life of this agent in this patient?

1b What is the expected initial plasma drug concentration of this dose of drug in this patient?

1c What is the volume of distribution of this agent in this patient?

1d How many hours following administration of the dose are required for 87.5% of the agent to be eliminated from the body?

1e If the dose were administered at a level of 10 mg/kg, how many hours following administration of the dose are required for 87.5% of the drug to be eliminated from the body?

1f What is the clearance of this agent in this patient?

2. The equation below represents the concentration of drug in the plasma following IV bolus administration of a 400 mg bolus dose. (Concentration is given in mcg/mL and time in hours)

$C_p = 9.46e^{-4,231t} + 6.25e^{-0.173t}$

2a What is the elimination half-life of this agent?

2b What is the initial concentration of drug in the plasma upon administration of the dose?

2c What is the concentration of drug in the plasma 4 hours after administration of the dose?

2d What is the apparent volume of distribution of the central compartment?